CHAPTER 3

Vector calculation and scalar product

Chapter overview

Objectives. Choose between projection, coordinates and angle; calculate a length; characterize Orthogonality and a locus.
Prerequisites. Vectors, Chasles· relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Annual benchmark. 6·15 October: 8 hours

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. For $A(1; 2)$ and $B(4; -2)$, calculate $\overrightarrow{AB}$.

2. Calculate the norm of $(3; -4)$.

3. Simplify $\overrightarrow{AB} + \overrightarrow{BC}$.

4. Develop $(a + b)^2$.

5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Find its cosine.

6. The midpoint of $A(-2; 1)$ and $B(4; 3)$?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Compare, on the same problem, coordinates and vector calculation without reference. Explain why your method uses the data most directly.

Search before formalizing

Build three M points such as $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for an AB segment of 6 cm. Conjecture the locus then produce evidence that includes A and B.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min. Keep a trace of a test, even if not completed. A conjecture must then be justified.

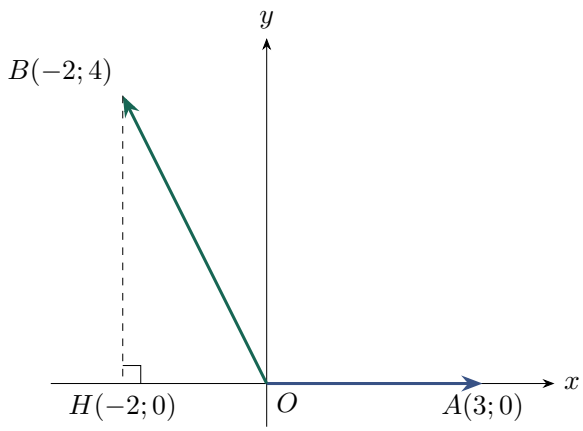
Projection and angle

Definition and results

For two non-zero vectors $\vec{u}, \vec{v}$, their scalar product is real $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$where: $\theta \in [0; \pi]$ is their angle not oriented. With a zero vector, the product is 0. If H is the projected of B on (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ value $OA \times OH$ when H is on the half-right [OA) and its opposite If not.

Status and justification. Definition; equivalence with projection results from cosine in the Rectangular triangles, with negative algebraic projection for an obtuse angle.

Example solved. For two vectors of norms 3 and 4 forming 60 °the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120 °, it's worth -6. If $A(3; 0)$ and $B(-2; 4)$, the projected is $H(-2; 0)$ and the product is -6.



Projected H is on the negative side of the horizontal axis. $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

Point of vigilance

The result is a number, not a vector. Right angle is 0, not 1.

Immediate application. Two vectors have norms 2 and 5 and form 90 °, then 180 °. Calculate products.

3.2 Coordinates, Norm and Orthogonality

Definition and results

In an orthonormous base, $\vec{u} = (x; y)$ and $\vec{v} = (x\prime; y\prime)$, $\vec{u} \cdot \vec{v} = xx\prime + yy\prime$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For a base $(i, \vec{j})$ orthonormous, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Status and justification. Demonstrated properties using bilinearity: develop $x\vec{i} + y\vec{j}) \cdot (x\prime\vec{i} + y\prime\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

Example solved. $\vec{u} = (2; 3)$, $\vec{v} = (3; -2)$: output 6 $- 6 = 0$. Both vectors are orthogonal. Their norm is $\overline{13}$. The product of (2; 3) with (1; 0) is worth 2: we find the abscisse.

Point of vigilance

The usual formula assumes an orthonormous basis. different properties.

Immediate application. For what real t the vectors (t; (2) and (3); -6) Are they orthogonal?

3.3 Vector calculations and identities

Definition and results

The scalar product is symmetrical and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. So $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. We're getting it. $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

Status and justification. Bilinearity accepted from the geometric definition; symmetry resulting Cosine. Identities are demonstrated by developing $\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

Example solved. If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. Deduct orthogonality without coordinates.

Point of vigilance

In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. Identities concern the squares of norms.

Immediate application. We give $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

3.4 Al-Kashi and choice of method

Definition and results

In an ABC triangle, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows calculate the third side from two sides and including the angle, or determine the cosine d-an angle from the three sides.

Status and justification. By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Developing its scalar square gives $BC^2 = AC^2 + AB^2 - 2AB \cdot AC$; introduce the expression with the cosine.

Example solved. $AB = 5,\ AC$ = 3 and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15$ = 19, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approaching the angle to the calculator if allowed.

Point of vigilance

The angle is between the two sides of the product term. The minus sign is essential.

Immediate application. A triangle a $AB = AC$ = 4 and $BC = 4\sqrt{2}$. Determine angle A.

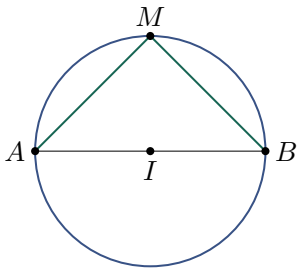
3.5. The AB diameter circle

Definition and results

For two separate points A and B of midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. All points M such as $\overrightarrow{MA} \cdot \overrightarrow{MB}$ = 0 is the centre circle I and radius circle $\dfrac{AB}{2}$, points A and B included.

Status and justification. Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. Cross-words cancel ; it remains $MI^2 - IA^2$. Like $IA = \dfrac{AB}{2}$, equal to zero is equivalent to the metric equation of the circle.

Example solved. $A(-2; 0)$ and $B(2; 0)$: I is the origin. The locus is $x^2 + y^2$ = 4. For $M(0; 1)$, the product is 1 $- 4 = -3$; for $M(0; 2)$, it's 0.



AB diameter circle: vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

Point of vigilance

The ends belong to the place even if "right triangle" is not suitable for a Degenerate triangle.

Immediate application. For $A(0;0)$, $B(6;0)$, determine M such as $\overrightarrow{MA}\cdot\overrightarrow{MB}=7$.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance. to 7 are a reserve of transfer, research or programming: the professor designates one. Exercise 8 resumes at a distance.

Fiscal year 3.1 · Direct calculations

In an orthonormous basis of the plan, vectors are considered $\vec{u}=(1;-2)$ and $\vec{v}=(4;3)$.

1. Calculate the scalar product $\vec{u}\cdot\vec{v}$, then the norms $\|\vec{u}\|$ and $\|\vec{v}\|$.
2. Calculate the coordinates of $\vec{u}+\vec{v}$ and deduct from it $\|\vec{u}+\vec{v}\|^2$.
3. To find this last result with the help of identity involving $\vec{u}\cdot\vec{v}$.

Exercise 3.2 · A right triangle?

The plan is equipped with an orthonormal coordinate system. $A(1;1)$, $B(4;2)$ and $C(0;4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.
2. Deducing that the triangle ABC is rectangle; specify at which top.
3. Calculate the lengths of the two perpendicular sides and then the triangle's area in diode units.

Exercise 3.3 · Length without reference

The Quadrilateral $ABCD$ is a parallelogram such as $AB=4$, $AD=3$ and $\widehat{BAD}=60°$. The lengths are expressed in the same unit.

1. Make a figure and calculate the scalar product $\overrightarrow{AB}\cdot\overrightarrow{AD}$.
2. Determine exact diagonal lengths $[AC]$ and $[BD]$. Explain how vectors corresponding to the diagonals expressed from $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
3. Indicate which diagonal is the longest and check that this result is consistent with your figure.

Exercise 3.4 · Find an angle

Two non-zero vectors $\vec{u}$ and $\vec{v}$ check $\|\vec{u}\|=2$, $\|\vec{v}\|=6$ and $\vec{u}\cdot\vec{v}=-6$. Note θ their angle no oriented, within 0 ° and 180°.

1. Determine the exact value of cos θ, then the measurement of θ.

2. Explain how the sign of the scalar product allowed predicting that the angle would be obtuse.

Exercise 3.5 · A variable location

Points A and B are distinct and fixed. I the midpoint of $[AB]$. For a real k set, we're looking All points M of the plan that verifies $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product according to MI and AB using Chasles' relationship and A scalar identity.
2. Describe precisely the desired set according to the value of k. For each case, indicate whether it is empty, reduced to a point or constituted by a circle; then give its characteristics.
3. Explain the role of whether a squared distance is positive or zero.

Exercise 3.6 · Force Work

An object moves in a straight line of A To B over a distance of 4 m. It is subjected to a force constant $\vec{F}$ 50 N. The work of this force is defined by $W = \vec{F} \cdot AB$ and expressed in Joules.

1. Calculate W when the angle is between $\vec{F}$ and $\overrightarrow{AB}$ value 60°.
2. Resume calculation for an angle of 120 °.
3. Compare the signs of the two works and explain, in each situation, whether the force favours the Displacement or opposing.

Fiscal year 3.7 · Contrapositive

The following statement, valid for two vectors of the plan, is to be demonstrated:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal} \quad .$$

1. State the contraposition of this involvement.
2. Demonstrate this contraposition by means of a scalar identity, and then conclude on the initial statement.

Fiscal year 3.8 · Deferred review

A triangle ABC checks $AB = 3$, AC = 4 and BC = 5. We are looking $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using any coordinates.

1. Calculate this scalar product by a formula involving the three lengths.
2. Find the result by determining the nature of the triangle. Quote the theorem used and its numerical hypothesis.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 250, questions a and b. Calculate scalar product in orthonormized base and detail both coordinates products.

CQFD, Exercise 11 at 250, Questions 1, 2 and 3. Use successively the coordinates, Chasles' relationship and the fact that I is the midpoint.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $(2;1)\cdot(3;-(4)$ is

   A. 2
   B. 10
   C. $(6;-4)$
   D. -2

2. The norm of (-3; 4) is

   A. 7
   B. 25
   C. 5
   D. 1

3. Two vectors of zero product are

   A. Equal
   B. Colinear
   C. Opposite
   D. orthogonal

4. Norms 2 and 3, angle 60 $^\circ$: product

   A. 6
   B. 3
   C. $\sqrt{3}$
   D. 0

5. $\|u-v\|^2$ value

   A. $\|u\|^2-\|v\|^2$
   B. $\|u\|^2+\|v\|^2$
   C. $\|u\|^2-2u\cdot v+\|v\|^2$
   D. $\|u\|^2+2u\cdot v+\|v\|^2$

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. An obtuse angle between non-zero vectors gives a product

   A. Positive
   B. None
   C. equal to 1
   D. Negative

2. On an orthonormous basis, $u\cdot i$ is

   A. abscisse de u
   B. the norm of u

C. The order of u

D. Still sucks

3. Location $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is

A. the mediator

B. the right AB

C. the circle of diameter AB

D. the midpoint alone

4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ is

A. -3

B. -6

C. 6

D. 9

5. For $AB = AC = 2$ and angle A right, BC value

A. 4

B. 2

C. 8

D. $2\sqrt{2}$

True or false?

1. The scalar product of two vectors is always positive.

2. The null vector is orthogonal to any vector.

3. $u \cdot v = 0$ implies that one of the vectors is zero.

Personal review

I distinguish norm and square norm; I choose a suitable formula; I check the reference; I write equivalence for a locus.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.

Corrected and conduct of chapter 3

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $(3; -4)$. *Action:* Subtract arrival less departure on two drawings.
2. $\sqrt{3^2 + (-4)^2} = 5$. *Action:* Review Pythagoras and distinguish norm and coordinates.
3. $\overrightarrow{AC}$. *Action:* Trace the compound displacement.
4. $a^2 + 2ab + b^2$. *Action:* Identify the cross-word.
5. $\dfrac{3}{5}$. *Action:* Name the sides in relation to the angle.
6. $(1; 2)$. *Action:* Measure the two coordinates separately.

Research activity

Take points on the AB diameter circle; demonstration by midpoint I: product $MI^2 - 9$.
Extremities are included because one of the vectors is zero.

Immediate applications

1. 0 then -10. Cosine is equal to 0 and -1.
2. $3t - 12 = 0$, therefore $t = 4$.
3. The square is worth $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.
4. $32 = 16 + 16 - 32\cos A$, therefore $\cos A = 0$ and $A = 90°$.
5. $MI^2 - 9 = 7$: centre circle (3; 0) and radius 4.

Exercises: approach and results

Fiscal year 3.1 · Direct calculations. Output $4 - 6 = -2$; norms $\sqrt{5}$ and 5; sum (5; 1) squared 26.
Control: $5 + 2(-2) + 25 = 26$.

Exercise 3.2 · A right triangle?. $\overrightarrow{AB} = (3; 1)$, $\overrightarrow{AC} = (-1; 3)$; product 0. $AB = AC = \sqrt{10}$, area $\dfrac{AB \times AC}{2}$ = 5 units of area.

Exercise 3.3 · Length without reference. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. Output $\dfrac{4 \times 3}{2} = 6$; $AC^2 = 16 + 9 + 12 = 37$ and $BD^2 = 16 + 9 - 12 = 13$.

Exercise 3.4 · Find an angle. $\cos\theta = -\dfrac{6}{12} = -\dfrac{1}{2}$; in [0; 180 °], the angle is 120°. The field ensures uniqueness.

Exercise 3.5 · A variable location. $MI^2 = k + \dfrac{AB^2}{4}$. If the second member is negative: together empty ; zero : the only point I ; positive : radius circle its square root. The discussion is about a distance to square.

Exercise 3.6 · Force work. $W = \dfrac{50 \times 4 \times 1}{2} = 100$ J. À 120°, $W = -100$ J. The sign Distinguishes aid for the displacement of opposition; it does not describe a negative distance.

Exercise 3.7 · Contrapositive. Direct "orthogonal" ownership implies equality of squares The contrast is exactly stated. You can also write 2 $\vec{u} \cdot \vec{v} \neq 0$.

Fiscal year 3.8 · Deferred review. Al-Kashi: $\dfrac{9 + 16 - 25}{2}$ = 0. Pythagoras Converse: Triangle right-angled at A, therefore zero product. Name the assumptions of each method.

CQFD, exercise 9 p. 250, questions a and b. a : $1 \times 3 + 2 \times (-1) = 1$. b : $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

CQFD, Exercise 11 at 250, Questions 1, 2 and 3. $\overrightarrow{AB} = (4; 6)$, $\overrightarrow{AC} = (-3; 4)$, $I(-1; -2)$ and $\overrightarrow{CI} = (5; -(1)$ Where $\overrightarrow{AB} \cdot \overrightarrow{AC} = 12$, $\overrightarrow{AB} \cdot \overrightarrow{CI}$ = 14. Chasles gives $\overrightarrow{AB} \cdot \overrightarrow{AI}$ = 12 + 14 = 26. controls: $\overrightarrow{AI}$ = (2; 3) gives 8 + 18 = 26; or $\overrightarrow{AI} = \dfrac{1}{2}\overrightarrow{AB}$ gives $\dfrac{1}{2}(16 + 36) = 26$.

MCQ: answers and reasons

MCQ 1.

1. A. $6 - 4 = 2$.

2. C. Root of 25.

3. D. Scalar criterion.

4. B. $\dfrac{2 \times 3}{2} = 3$.

5. C. Developing the scalar square.

MCQ 2.

1. D. Negative Cosine.

2. A. Projection coordinates.

3. C. $MI^2 = \dfrac{AB^2}{4}$.

4. B. Bilinearity: 2($-1)3$.

5. D. $BC^2 = 8$.

True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.

2. True: its product with any vector is 0.

3. False: (1; 0) and (0; 1) are non-zero and orthogonal.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.